

The law of small numbers

When statistics and psychology go to war

Humans have innate biases when it comes to probability, leading to poor decision-making. **Itamar Shatz** looks at a bias that affects disinformation-spreaders and statisticians alike

You just tossed a fair coin twice, and it landed on heads both times. If you toss it again, do you think that it will land on heads or on tails? As we will see, the answer that people might give to this question can be surprising, and has important implications for various high-stakes issues, such as flawed research, social stereotyping, and vaccine misinformation.

From a statistical perspective, the answer to the above question is simple: each toss is independent of the others, since the coin cannot remember past tosses, so it does not matter that it previously landed on heads. Accordingly, the likelihood that the coin will land on heads next is equal to the likelihood that it will land on tails. However, from a psychological perspective, the answer is more complicated than that.

To understand why that is, it helps to first consider the *law of large numbers*, a concept in probability theory which says that as the size of a sample increases, it will generally become more representative of the parent population from which it is drawn.¹ In the case of coin tosses, this means that we would expect long sequences of tosses to have a roughly equal (1:1) ratio of heads to tails (H:T). Conversely, although short sequences of tosses (e.g., 5) can also have an equal H:T ratio, they are more likely than long sequences to have a distinctly non-equal ratio (e.g., 4:1).

As its name suggests, the law of large numbers applies to large samples; the larger the better. But, paradoxically, people often expect the law of large numbers to apply to

small samples too. This irrational expectation is the *law of small numbers*, a cognitive bias that causes people to assume that small samples are as representative of their parent population as large samples are.^{2,3} This can cause people to expect short sequences of coin tosses to have a roughly equal H:T ratio, similarly to long sequences.

Figure 1 illustrates these two “laws” and how they relate to each other.

Essentially, the law of small numbers represents the psychological (rather than statistical) expectation of *local representativeness*. This means that people

expect the characteristics of the (global) population to be represented locally, in all of its parts, including in small ones.^{2,3} This expectation, which relates to the view of chance as a fair and self-correcting process, is further illustrated in Figure 2.

Crucially, belief in the law of small numbers can influence statistical reasoning about many real-life events beyond coin tosses. For example, it can lead to the well-known *gambler’s fallacy*, when it causes people to assume that the colour black is “due” in a roulette roll, if the ball previously fell on red several times in a row.²⁻⁴ Similarly,

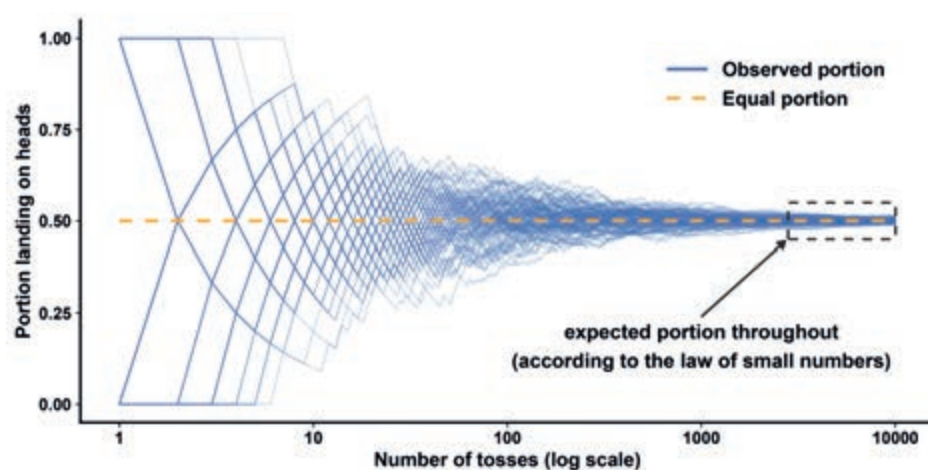


Figure 1: The portion of times a coin lands on heads, as a function of the number of tosses (from 1 to 10,000). Each line represents 1 out of 100 random sequences of tosses; darker shading denotes an overlap of multiple lines. In the initial tosses, the portion often fluctuates substantially away from an equal portion of heads and tails. Then, as the number of tosses increases, the portion approaches equality and the fluctuations become smaller, in line with the law of large numbers. However, believers in the law of small numbers expect the ratio to be near equality and with small fluctuations (e.g., like it is after 1,000 or even 10,000 tosses), even during the first few tosses.



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(A) Observed tosses:



(B) Expected tosses, according to the law of small numbers:



(C) Sorted tosses, both observed and expected:



Figure 2: (A) shows an observed sequence of 50 random coin tosses, landing on either [H]eads or [T]ails, with marking for cases where the coin lands on the same side more than twice in a row. (B) shows a sequence of tosses that believers in the law of small numbers might expect, where the coin rarely lands on the same side more than once in a row, and never more than twice. (C) shows the identical sorted tosses of the previous sequences, with heads on one side and tails on the other, demonstrating that the *global* H:T ratio of both sequences is the same, despite *local* differences between the sequences.

it can cause people to assume that if several girls have been born in a row to a family, then a boy is “due” next.

In addition, the law of small numbers can influence people’s reasoning about other types of outcomes. For instance, it can cause people to stereotype social groups based on factors such as nationality or gender, by overgeneralising the behaviour of a single group member, due to the expectation that this person will be highly representative of their entire group. Similarly, the law of small numbers can cause researchers to overestimate how representative small samples are, and consequently to study underpowered samples and be overly confident in findings from those samples.² A classic example of this appears in the following question, which you can try to answer yourself.

Suppose you have run an experiment on 20 subjects, and have obtained a significant result which confirms your theory ($z = 2.23$, $p < 0.05$, two-tailed). You now have cause to run an additional group of 10 subjects. What do you think the probability is that the results will be significant, by a one-tailed test, separately for this group?

This question was posed to 84 people at meetings of the Mathematical Psychology Group and of the American Psychological Association, as part of the original research on the law of small numbers, which was published in 1971.² If your answer to the question was around 0.85, then you

are in good company, since this was the median answer that respondents gave. Unfortunately, this answer is wrong, since the actual probability is around 0.48, as explained in the original paper. Most of the respondents got this wrong despite their relevant expertise, since only nine of them (11%) came close to the correct solution, with an answer in the range of 0.40–0.60, a discrepancy that demonstrates the powerful influence of this bias.

Finally, another important example of the law of small numbers can be found in vaccine misinformation, which has surged during the Covid pandemic. This is illustrated in the following argument, which is frequently found on social media in various forms:

My wife and I both caught Covid despite being vaccinated, so the government is lying when it says that the vaccine is 80% effective.

Here, the poster incorrectly assumes that if the vaccine is 80% effective in the total population, then this effectiveness rate should be represented locally, in a sample of just two people. Moreover, this argument reflects many other issues that are common in statistical reasoning – beyond the law of small numbers – such as the tendency to ignore the influence of personal factors (e.g., age) on the likelihood of catching Covid, which could mean that these two people are more likely to catch Covid than average members of the population. The flawed reasoning in this case can cause

serious problems, such as misinformation dissemination and vaccine refusal, sometimes with life-and-death consequences.

Overall, the law of small numbers can substantially interfere with statistical reasoning in many situations, by causing people to overestimate how representative small samples are of their parent population. Accordingly, it serves as a good reminder that people are not *Homo economicus* – perfectly rational beings capable of flawless reasoning – but rather are fallible creatures, prone to various cognitive biases.

Because of the issues that the law of small numbers can cause for judgement and decision-making, and because of the potential consequences of those issues, it is important to mitigate this bias, in both yourself and others. To do so, you should understand this bias, including when and how it can influence people, and increase awareness of it. If you then encounter a situation where this bias might play a role, for instance because you are looking at a small sample of people, then you can actively consider how representative the sample in question, and *why*. When engaging in this debiasing, you can consider factors such as how big your sample is, whether it was selected in a random manner, and how much variance is expected in your variable of interest. Asking and answering this explicitly, by properly analysing evidence rather than relying on intuition alone, can reduce the likelihood that you will fall for the law of small numbers, which in turn can lead to better reasoning. ■

Code availability

The code used to generate the figures is available at osf.io/fa2qv

References

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